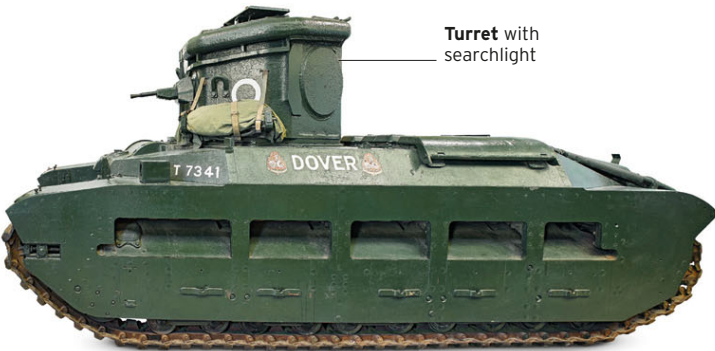


Engineering and Specialized Vehicles

After the failed Dieppe Raid of 1942 exposed the difficulty of landing vehicles during an amphibious invasion, Allied commanders knew that getting tanks across the beaches of France would be a challenge. The job of developing suitable vehicles was given to Percy Hobart, the commander of the British 79th Armoured Division. Known as “Hobart’s Funnies,” these vehicles were based on tank hulls, which gave them similar mobility and protection, and made logistics easier. They were used in northwest Europe, Italy, and the Far East.



Turret with searchlight

△ Matilda CDL
Date 1940 Country UK
Weight 29.7 tons (26.9 tonnes)
Engine 2 x AEC 6 cylinder diesel, 95 hp each
Main armament None

The Canal Defence Light (CDL) was an attempt to dazzle the enemy during night fighting. The turret of the Matilda contained a 13-million candle power searchlight that flickered at a frequency that increased the blinding effect.

▷ Valentine Bridgelayner
Date 1943 Country UK
Weight 20 tons (19.9 tonnes)
Engine AEC A189 gasoline, 135 hp
Main armament None

The first bridgelaying tanks were developed at the end of World War I, but it was not until World War II that they were used. The Scissors Bridge shown here could span a 30 ft (9.2 m) gap and support 33 ton (30 tonne) vehicles.



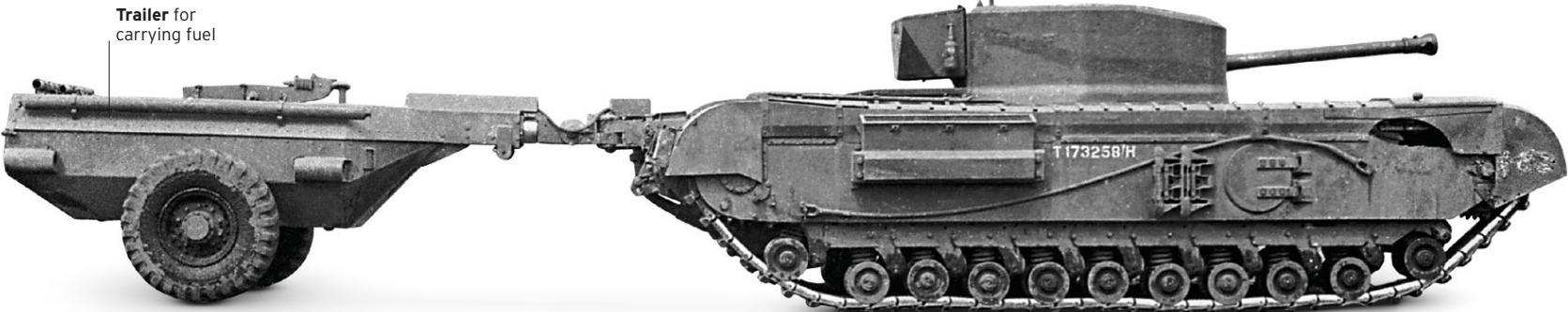
Mine housed in vehicle body

△ Goliath tracked mine
Date 1943 Country Germany
Weight 0.5 tons (0.4 tonnes)
Engine Zundapp SZ7 gasoline, 12.5 hp
Main armament 100 kg (220 lb) explosive

Just 5.3 ft (1.63 m) long and 2 ft (0.62 m) tall, the Goliath was effectively a small bomb. It was remotely controlled by a 2,130 ft (650 m)-long wire, which allowed its operator to remain in cover. It was intended to be used against fortifications or to clear minefields, but was vulnerable to small arms fire and rough terrain.

▽ Churchill Crocodile
Date 1943 Country UK
Weight 44.8 tons (40.6 tonnes)
Engine Bedford Twin-Six gasoline, 350 hp
Main armament Flamethrower, 75 mm QF gun

A flamethrower is extremely effective against fortifications, and mounting one onto a tank enables the latter to survive enemy fire as it closes in. The Churchill Crocodile was one such vehicle; a fully operational gun tank with a trailer for carrying fuel. Crocodiles attracted heavy enemy fire, but their presence often persuaded German forces to surrender.



Trailer for carrying fuel